Ancestor relevance gate

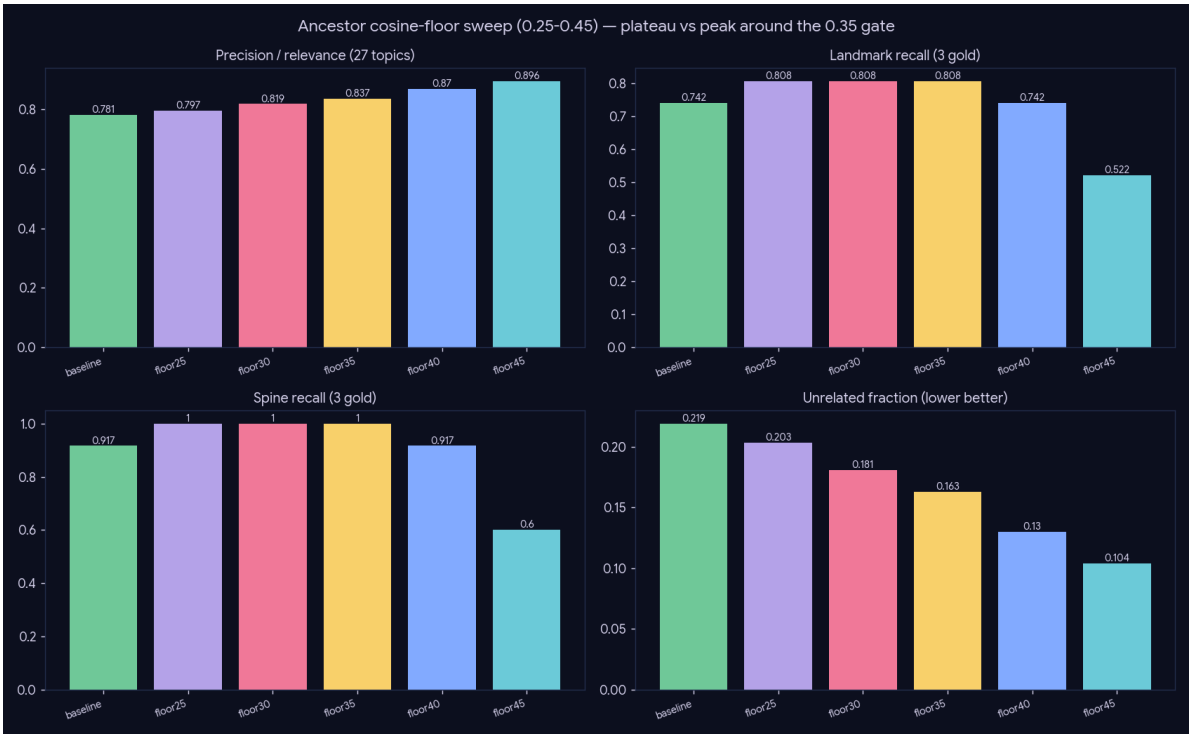
GraphIV · retrieval · plugging the noise mechanism

What we did

`02-02` found ~22% of landmarks off-topic AND carrying high in-degree (5.4) — famous cross-field hubs (GPT-3 / StyleGAN under "diffusion"). They enter through one mechanism: `augment_with_ancestors` → `store.ancestors` pulls the top-120 *most-cited* one-hop-back papers (ranked by citer count *before* cosine), so fame beats relevance. We gate that injection by a cosine floor, sensitivity-swept over a **0.25–0.45 grid** (step 0.05 — the original 0.35 came from a coarse search, so the ship decision needed plateau-vs-peak), and measure **both** sides: **precision** (LLM `relevance_rate` over 27 topics, reusing the `02-02` cache) and **recall** (`landmark_recall` + `spine_recall` over 3 gold topics). Only the injected ancestors are gated; the core pool and cosine-gated `frontier_pull` are untouched.

Results

variant	precision (27)	unrelated	landmark rec. (3)	spine rec. (3)
baseline	0.781	0.219	0.742	0.917
floor25	0.797	0.203	0.808	1.0
floor30	0.819	0.181	0.808	1.0
floor35	0.837	0.163	0.808	1.0
floor40	0.87	0.13	0.742	0.917
floor45	0.896	0.104	0.522	0.6



Conclusion

Recall sits on a flat plateau across 0.25–0.35 — landmark recall 0.808 and spine 1.0 are *identical* at all three floors while precision rises monotonically (0.797 → 0.819 → 0.837; baseline 0.781/0.742). The gains are not tuned to a magic number: any gentle floor beats no floor on both metrics, and **0.35 is the max-precision edge of the plateau** — the principled ship point. **The cliff starts at 0.40** (recall falls back to baseline) and **0.45 collapses it** (0.522, spine 0.6): those floors cut genuine low-cosine foundational papers (VAE / Sohl-Dickstein sit ∼0.4–0.5). An earlier pass also tested a modal-category gate at 0.45; it added ∼nothing and is dropped. **Verdict: shipped** — `augment_with_ancestors(min_cosine=0.35)` is the production default (`pipeline.py` + the harness recipe); 0 remains the ablation arm. **Caveat:** recall is n= 3 gold, precision n= 27; the three-floor plateau plus the mechanism make the no-recall-loss claim robust, but a wider gold set would harden it further.